B. Reshmasri
ECE-A
A24126512010

PYTHON

Assignment-2

1. Student Academic Management System

A college wants to digitally manage student academic records, including marks, CGPA, age calculation, and fee balance. Create a class named student with the following attributes: Student ID, Student Name, Marks in 3 subjects, Date of Birth, Fee paid. Create methods to: calculate average marks and CGPA, calculate age of the student, calculate fee balance. Create another class named college with the following attributes: College code, College name, Location. Create methods to: register a student, display complete student academic details.

Assume a college registers 3 students and wants to generate a student academic report. Write a Python program to display: college details, student details, CGPA, Age, Fee balance

Code/Program:

```
Class Student:
    total-fee = 1,00,000
        def __init__ (self, id, sname, m1, m2, m3, dob,
                                        fee-paid):

        self.id = id
        self.sname = sname
        self.m1 = m1
        self.m2 = m2
        self.m3 = m3
        self.dob = dob
        self.fee-paid = fee-paid
        def calculate_average (self):
        avg = (self.m1 + self.m2 + self.m3)/3
        print ("Average marks : ", avg)
```

```
def calculate_cgpa(self):
    avg = (self.m1 + self.m2 + self.m3)/3
    cgpa = avg/10
    print("CGPA : ", cgpa)
def calculate_age(self):
    age = 2026- self.dob
    print("Age:", age)
def fee_balance(self):
    bal = student.total_fee - self.fee_paid
    print("Fee balance: ", bal)
def display_student(self):
    print("Student ID : ", self.id)
    print("Student Name: ", self.sname)
    print("Marks : ", self.m1, self.m2, self.m3)
    print("Fee paid : ", self.fee_paid)
    self.Calculate_average()
    self.calculate_cgpa()
    self.calculate_age()
    self.fee_balance()
    print("  ")

class college:
    def __init__(self, code, name, location):
        self.code = code
        self.name = name
        self.location = location
        self.students = [ ]
    def register_student(self, student):
        self.students.append(student).
```

```
def display_details(self):
    print("college code: ", self.code)
    print("College name:", self.name)
    print("Location:", self.location)
    print("In Student Academic Report")
    print("        ")
    for s in self.students:
        s.display_student()

college = college("ANIL", "ANITS college", "Visakhapatnam")
s1 = student(10, "Reshmasri", 90, 95, 92, 2007, 50000)
s2 = student(87, "Ramya", 85, 81, 95, 2006, 32000)
s3 = student(35, "Sita", 95, 96, 99, 2001, 42000)
college.register_student(s1)
college.register_student(s2)
college.register_student(s3)
college.display_details()
```

```
College Code: ANITS
College Name: ANITS College
Location: Vizag

Student Academic Report

Student ID: 1
Student Name: Reshma
Marks: 80 76 82
Fee Paid: 60000
Average marks: 79.33333333333333
CGPA: 7.933333333333333
Age: 17
Fee balance: 40000

Student ID: 2
Student Name: Ramya
Marks: 75 89 90
Fee Paid: 50000
Average marks: 84.66666666666667
CGPA: 8.466666666666667
Age: 18
Fee balance: 50000

Student ID: 3
Student Name: Sita
Marks: 85 96 99
Fee Paid: 70000
Average marks: 93.33333333333333
CGPA: 9.333333333333332
Age: 19
Fee balance: 30000
```